

The Moving Phantom: a Khronometric MOND Completion That Passes Local Gravity and Fails Cosmology as Built, and Postquantum Classical Gravity Tested as the Origin of $a_0 \propto \sqrt{\Lambda}$

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Abstract

The relation $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fits galaxy data (measured $\kappa = 0.465 \pm 0.076$) but supplies no mechanism. This note reports two mechanism tests run in September 2026. Each rests on a committed script with a mutation control, re-run for this note; the completion's algebra is also certified in Lean 4 (62 theorems, zero **sorry**). (i) Postquantum classical gravity predicts $a_0 \propto \sqrt{\Lambda}$ from stochastic metric noise (Oppenheim & Russo 2024). Its published coefficient is $\kappa = 1.30\text{--}1.45$, at least 11σ above the measured value. Its linear term gives a lensing radial-acceleration slope of 0 against the measured 0.537 ± 0.026 . Around the classical saddle the fluctuation action contains no matter, so any rectified density is uniform: the slope is $-\frac{1}{2}$ (39σ), and a density large enough to act in galaxies would be $\geq 4 \times 10^4$ times the cosmic dark matter. A phantom made of gravitational-wave energy lenses at 0.75 of its dynamics (10.6σ). (ii) In the programme's covariant candidate C-H the MOND sector carries no momentum of its own, so the baryons cannot generate a phantom: at linear order the non-baryonic density seen by the leaf curvature is frozen. Adding the khronometric α - and λ -terms with a monotone phantom law gives C-H/K. Its phantom forms and follows moving sources. Its modes are free of ghosts and tachyons at leading order. Its Solar-System metric is that of general relativity at first post-Newtonian order ($\gamma = \beta = 1$ derived), and it leaves the SPARC fit unchanged. A moving source's phantom is amplified by $1/(1 - v_{\parallel}^2/c_s^2)$. Cosmologically C-H/K fails as built. With the cold fluid the CMB requires, $\sigma_8 = 18\text{--}27$. On FRW it is low-energy Hořava gravity, whose Planck-era bounds exclude the λ -window that tracking needs. A switch keyed to the leaf curvature restores linear growth, but by Gauss's law it makes isolated galaxies weigh only their baryons beyond its edge. It also leaves the cosmic web Newtonian, and the field that web exerts on isolated lenses ($0.015 a_0$) is about $200\times$ what KiDS-1000 isolated lensing allows: $\Delta\chi^2 = +404/+415$ even with a free two-halo term, and $+47/+52$ inside 0.3 Mpc, where isolation is certain. Neither result derives κ .

1 The question

The programme's central relation is

$$a_0 = \kappa c \sqrt{G \rho_\Lambda}, \quad \kappa = \frac{1}{2} \iff a_0 = \frac{c H_\Lambda}{\sqrt{32\pi/3}}, \quad H_\Lambda = H_0 \sqrt{\Omega_\Lambda}. \quad (1)$$

Wherever a_0 enters, the lanes below are run on two footings: canonical $a_0 = 9.3619 \times 10^{-11} \text{ m s}^{-2}$ (ρ_Λ from Planck) and alternative $1.1279 \times 10^{-10} \text{ m s}^{-2}$ (ρ_{crit} in place of ρ_Λ ; ratio $1/\sqrt{\Omega_\Lambda} = 1.2048$). The coefficient is fitted: $\kappa = 0.465 \pm 0.076$ from the baryonic Tully–Fisher intercept and 0.55 ± 0.17 from a distance-free shape statistic. In the local MOND action classes examined, κ is a free zero mode [Z26a]. The relation therefore states a coincidence, and a mechanism has to supply three things: (a) the tie of a_0 to Λ , including its coefficient; (b) a relativistic completion whose phantom lenses as it attracts, moves with its sources, and passes the Solar-System and gravitational-wave tests; (c) a cosmology. Section 2 tests the one published first-principles route to (a). Sections 3–5 build and test a completion for (b) and (c). Section 7 states what the results do and do not change for the relation itself.

2 Postquantum classical gravity as the origin of $a_0 \propto \sqrt{\Lambda}$

In postquantum classical gravity the metric stays classical and must diffuse when it couples to quantum matter [Opp23]. Oppenheim & Russo [OR24] argue that the diffusion produces a MOND-scale term with $a_0 \sim \sqrt{\Lambda}$. Hertzberg & Loeb [HL24] reply that the Gaussian mean equation is only $\nabla^4 \Phi = 4\pi G \nabla^2 \rho$, so the linear term is not a free solution and is not MONDian. Classical metric noise is not suppressed by $\hbar$, so it escapes this programme's no-go for quantum graviton noise, which falls short by $\sim 10^{-124}$. It was therefore the one external claim of the relation's scaling that had not been priced (lanes L313, L314).

The published coefficient (L313, section A). Reading OR24's linear term as $a_0 := |\gamma_1| c^2/2$, with r_{max} the de Sitter radius and completely positive $\beta < 0$, gives $\kappa \in (1.3025, 1.4472)$. That is $\geq 11.0\sigma$ above 0.465 ± 0.076

and $\geq 4.4\sigma$ above 0.55 ± 0.17 . Obtaining $\kappa = \frac{1}{2}$ needs $\beta = 0.2353$. That value lies inside OR24's positivity range ($\beta \leq 1/3$) but outside complete positivity. With $r_{\max} = c/H_0$ instead, $\kappa \in (1.078, 1.198)$, still 8.1σ high. Used as a Tully–Fisher normaliser ($v^4 = 4GMa_g$ at the flattest point), $\beta \rightarrow -\infty$ gives $a_0 = 2cH_\Lambda$, i.e. $\kappa = \sqrt{32\pi/3} = 5.789$. That is the de Sitter–Unruh value [Mil99], which this programme's fits exclude. Independently of the coefficient, the $\gamma_1 r$ term is a constant acceleration. It gives a lensing radial-acceleration-relation (RAR) slope $d \ln g_{\text{obs}}/d \ln g_{\text{bar}} \rightarrow 0$ against the KiDS-1000 value 0.537 ± 0.026 [Bro21] (20.4σ), which is Hertzberg & Loeb's point, measured. Their second point is also checked: $\nabla^4 r = -8\pi\delta^3$, so the linear term needs a point source.

Does a rectified phantom lens as it attracts? (L313, section B). We average the exact second-order Einstein tensor over zero-mean metric noise in the short-wavelength (Isaacson) regime. The mean metric then obeys the linear equations with an effective source, and the ratio of lensing to dynamics is fixed by $\epsilon = (p_r + 2p_t)/\rho_{\text{eff}}$ alone. The KiDS-1000 gate is $|\epsilon| < 0.0489$ at 1σ . The averaged tensors are rotation scalars, so every bound holds for any directional structure of the noise.

- Noise that obeys GR's constraints can only consist of transverse-traceless waves. It gives $\epsilon = +1$ and lensing/dynamics = 0.750, excluded at 10.6σ in every averaging convention. This covers any graviton bath.
- Constraint-violating static scalar noise, the case CQ gravity uniquely allows, is treated under the Einstein–Langevin convention $\langle G_{\mu\nu} \rangle$. It has minimum $|\epsilon| = (\sqrt{53} - 7)/4 = 0.0700$ at $\phi = u_*\psi$, $u_* = (\sqrt{53} - 5)/4$. The phantom must lens at least 3.8% more than it attracts: 1.6σ today, not excluded. OR24's own ansatz $\phi = \psi$ gives $\epsilon = -1/9$, a 6.25% excess (2.6σ).
- Pure lapse noise rectifies to repulsion. So does OR24's white-in-time limit ($\omega \gg ck$).
- The averaging convention is an $O(1)$ systematic for constraint-violating noise. The minimum $|\epsilon|$ is 0.070, 0.10 and 0 for $\langle G_{\mu\nu} \rangle$, $\langle G^\mu{}_\nu \rangle$ and $\langle G^{\mu\nu} \rangle$, and the densitised average carries no active mass at all. CQ gravity has to say which tensor density its noise is unbiased in.
- A two-component spectrum, the static optimum plus a fast anti-correlated part at weight $w_0 = 0.170$, reaches $\epsilon = 0$. Lensing alone therefore does not close the class.

To be the deep-MOND phantom, the metric's rms acceleration must exceed the mean field by $\sqrt{2/C_{\text{act}}} c/v_f \approx 1135\times$ at the MOND radius. Nothing in the theory fixes that amplitude.

The homogeneous-noise pincer (L314). Expand the Newtonian CQ action $-c_0 \int (\nabla^2 \Phi - 4\pi Gm)^2$ around the saddle $\nabla^2 \Phi_{\text{cl}} = 4\pi Gm$. The fluctuation action is $-c_0(1 - \beta)G^{-2} \int (\nabla^2 \delta\Phi)^2$, which contains no $m(\mathbf{x})$. The noise statistics are therefore translation-invariant whatever the matter does, and the rectified density is uniform in every convention. A uniform density gives $v^2 = GM/r + \frac{4\pi}{3}G\rho r^2$ and a lensing-RAR slope $\rightarrow -\frac{1}{2}$ against $+0.537$ (39.4σ). It also fails on magnitude. A uniform density large enough to be a galaxy's phantom at its MOND radius exceeds the whole cosmic dark-matter density $\Omega_{\text{dm}}\rho_{\text{crit}}$ by $\geq 4.1 \times 10^4$, and by ≈ 415 at ten MOND radii (most favourable galaxy mass and footing). The covariant weight $\sqrt{-g}D(g)$ makes the variance source-dependent only at $O(v^2/c^2) = 3.9 \times 10^{-7}$, whereas the phantom must vary by $\sim 10^4$ across a galaxy. *Verdict:* published CQ gravity, with constant couplings D_0, β , is closed as the source of the MOND phantom and of κ . It survives only with a diffusion kernel $D_0(\mathbf{x})$ tuned to track the phantom, which writes the law in by hand. *Scope:* Oppenheim, Panella & Pontzen [OPP26] find that metric diffusion at the end of inflation can produce a phantom component behaving as cold dark matter. That is a dark-matter route, observationally Λ CDM-like, not the a_0 - Λ law tested here. Nothing in L313/L314 bears on it.

3 The moving-source gate

C-H is the programme's covariant candidate for the MOND sector (repository file `qwen_claude_field_theory/closure_2026/g03_covariant_action_2026/ACTION.md`). It is built on a dynamical clock τ with unit normal $n_\mu = -\partial_\mu \tau / \sqrt{X}$, lapse $N = X^{-1/2}$, acceleration $a_\mu = D_\mu \ln N$ and expansion $K = \nabla_\mu n^\mu$:

$$I_{\text{CH}} = \frac{c^3}{16\pi G} \int \sqrt{-g} \left\{ R - 2\Lambda + 2|DU - a|^2 + 2\alpha^2 q(|DW_b|^2/\alpha^2) + \int_0^b dz L(\partial_z W - \Delta_h W) + \lambda_0(W_0 - U) \right\} + I_{\text{m}}, \quad (2)$$

Here $\alpha = a_0/c^2$, $W_b = e^{b\Delta_h} U$ is the heat-filtered auxiliary field ($b = \xi^2/2$), and $q'(s^2) = \nu(s) - 1$ carries the interpolating function. Its static branch reproduces the programme's screened (heat-filtered) AQUAL functional, including the Cassini floor and lensing = dynamics at leading static order. The programme's earlier clock-host

candidate [Z26b] has a MOND scalar with the same momentum-free structure; its khronon c_2 channel is the escape taken up below.

The frozen phantom (L330, 5/5). Every term of C-H’s modification is independent of the shift and of all time derivatives (symbolic check in unitary gauge). Its momentum constraint is therefore GR’s, with matter momentum only. Linearised around flat space with GR’s kinetic structure, the divergence of the momentum constraint is $\frac{1}{2}\partial_t R^{(3)}$, with every shift term cancelling. So $\partial_t[R^{(3)} - 16\pi G\rho_m] = 0$ for conserved matter: the non-baryonic density seen by the leaf curvature is frozen. The control $\lambda \neq 1$ breaks the identity. The baryons therefore cannot generate a phantom, and a galaxy assembling from near-uniform data forms none. The same fact, $\partial L_{\text{extra}}/\partial N^i = 0$, is what gives $\alpha_1 = \alpha_2 = 0$. *Scope.* The identity is linear around a phantom-free background, where the transport of a phantom by the aether’s own flow is a second-order term. In the nonlinear post-Newtonian treatment of a khronon theory with the same shift-free structure, Blanchet & Skordis [BS24], following Flanagan [Fla23], find that the extra sector’s energy density is a conserved fluid carried by the aether’s flow $v = -\nabla\sigma$ (their eqs. 27–30). Lane BSK1 (M2) re-derives this from their equations: the transport term is second order about flat space and first order about a galaxy. Read that way, the frozen identity says the phantom is conserved rather than generated, which keeps the formation conclusion. Whether an existing phantom follows a moving source then depends on the aether’s own dynamics, which L330 did not compute for C-H. L330’s further results assume linear order with the aether at rest, and hold only there: a $10^7 M_\odot$ satellite at 150 km s^{-1} leaving its MOND radius in 0.8 Myr, and lensing at about half the dynamics ($g_{\text{lens}}/g_{\text{dyn}} = 0.51\text{--}0.65$ across KiDS accelerations). The same restriction applies to L330’s conclusion that preferred-frame safety and a co-moving phantom exclude each other.

The khronon’s λ -channel carries it (L333 9/9, KM1 7/7). Add the khronometric term $-c_2 K^2$ (so $\lambda - 1 = c_2$). At $\lambda = 1$, in linear theory with the aether at rest, a moving phantom is absent from both gauge-invariant potentials, which reproduces the frozen phantom. At $\lambda = 1 + \varepsilon$ the shift carries a $1/\varepsilon$ piece. It cancels in the Bardeen potentials Φ_B, Ψ_B , so matter and light see the boosted static field to $O(w^2/c^2)$, with lensing = dynamics and α_1, α_2 unchanged. The price is in the aether’s own acceleration, which is the MOND sector’s input. It is distorted by $D \cos^2 \theta$ with $D = 2w^2/(\varepsilon c^2)$. At the Local Group’s 620 km s^{-1} through the CMB frame and $\varepsilon \sim (1\text{--}2.5) \times 10^{-5}$, this gives $D/3 = 0.29\text{--}0.11$. So either a_0 tracks each galaxy’s CMB-frame speed, or $c_2 \gtrsim 10^{-3}$, in which case α_2 forces $c_{14} \lesssim 8 \times 10^{-7}$. Massive clusters are nonlinear here: the khronon’s energy is 8–55% of the phantom.

4 C-H/K: the filtered-khronon completion

$$I_{\text{CHK}} = I_{\text{CH}} + \frac{c^3}{16\pi G} \int \sqrt{-g} [\alpha_c a_\mu a^\mu - c_2 K^2], \quad (3)$$

These are the Blas–Pujolàs–Sibiryakov α - and λ -terms [BPS10, BPS11], with $\beta = 0$ so that gravitational waves travel at exactly c . The C-H kernel is built from a *monotone* phantom law ν_{mono} . Below the phantom peak of ν_{RAR} ($y_p = 2.54$, $h_p = 0.648 a_0$) it equals ν_{RAR} ; above the peak the phantom rises slowly ($h' = 0.05 h_p/(y + y_p)$). The largest difference in $\log g_{\text{obs}}$ is 0.0104 dex. On static slices $K = 0$, so every static result of C-H carries over, and α_c only renormalises G by $O(10^{-9})$. Khronon-based relativistic MOND has a literature [BM11, San11, BB15, Fla23, BS24, BS25]. In C-H/K the MOND law lives in C-H’s filtered auxiliary sector, and the khronon terms serve only as a momentum channel and a regulator. “First on the record” below refers to this programme’s record.

Local gates (L340, 11/11).

- *Tracking.* With the c_2 channel the $\omega \rightarrow 0$ response is the static MOND solution, $\psi/\psi_N = -2(1 + C)/(\alpha_c(1 + C) - 2)$, with no frozen mode. The control $c_2 = 0$ gives $\det M(0) = 0$ and a Newtonian response.
- *Health.* Every kernel cell (y , direction) is healthy at leading order ($\omega^2 > 0$, positive energy), with mode speed $c_s^2/c^2 = c_2/[C(2 + 3c_2)]$. All 243 of 243 scanned cells with a negative constitutive coefficient are ghosts or tachyons. A monotone phantom law is therefore *required*: ν_{RAR} itself, with minimum $C_L = -0.032$, cannot be completed this way.
- *Negative-phantom lobes.* At $O(\Phi/c^2)$ the clock inertia crosses zero in lobes where $\rho_{\text{ph}} < 0$. The $\alpha_c a^2$ term removes this once $\alpha_c \gg G|\rho_{\text{ph}}|\xi^2/c^2 \sim 10^{-18}\text{--}10^{-13}$.
- *Galaxies and the Solar System.* On SPARC [LMS16], $\Delta\chi^2 = -0.8/+0.5$ on 18 bins. The Solar-System floors (0.031/0.045 pc canonical) are identical to ν_{RAR} ’s, because the heat filter keeps $y < 2.54$ inside ξ . Without the

filter the monotone tail exceeds the ephemeris bound by 1.9×10^4 . The filter thereby breaks the programme's earlier Cassini-versus-ghost pincer.

- *Window.* $\alpha_c \in (9.6 \times 10^{-14}, 3.2 \times 10^{-9})$ and $c_2 \in (7.3 \times 10^{-3}, 0.067)$. The lower edge of c_2 is the tracking condition ($c_s \geq 3 \times 600 \text{ km s}^{-1}$ where $C \leq 100$). The upper edge (BBN) is superseded in §5.

The moving-source law (KM2, 4/4). In the same block, a source moving at v through the preferred frame has its potential multiplied by

$$R = 1 + \frac{C}{1+C} \frac{v_{\parallel}^2}{c_s^2 - v_{\parallel}^2} \quad (4)$$

(accurate to 1%). The phantom is *amplified* by $1/(1 - v_{\parallel}^2/c_s^2)$, a resonance rather than a lag, and diverges at the mode speed. At $c_2 = 7.3 \times 10^{-3}$ the effect is $\leq 0.3\%$ in galaxy bodies, $\leq 4\%$ in Mpc-scale lensing outskirts and $\leq 5\%$ in cluster lensing. All are below current reach.

First post-Newtonian order (KM3, 5/5). From the khronometric field equations, $\gamma_{\text{PPN}} = 1$ follows from the static linear solution. $\beta_{\text{PPN}} = 1$ follows from the static second-order equations; the control $\alpha_c = 0$ reproduces isotropic Schwarzschild, and a $\gamma = 2$ exterior gives $\beta = 3/2$. The preferred-frame parameters are $\alpha_1 = -4\alpha_c$ and $\alpha_2 \simeq -\alpha_c/2$, with the Nordtvedt parameter $\eta_N = \frac{11}{3}\alpha_c \leq 1.2 \times 10^{-8}$ across the window. $\alpha_3 = \zeta_i = \xi = 0$ by the structure of the action. The filtered MOND remainder enters 1PN only as (floor-bounded anomaly) $\times U \sim 10^{-9}$. *In the Solar System C-H/K's 1PN metric is GR's.*

5 Cosmology

Linear growth (L341, 7/7). With the cold fluid the CMB requires, C-H/K's MOND sector acts on the linear density field. It gives $\sigma_8 = 17.7\text{--}27.5$ across the c_2 window, both footings and both field arguments (observed 0.811). Finite tracking does not help: $c_2 = 10^{-6}$ still leaves $\sigma_8 > 1.3$. The one floor built from the construction's own fields is the clock's expansion K . In quadratic form it is unhealthy in deep MOND. In quartic form it is healthy on FRW and returns $\sigma_8 = 0.810$. The foliation is, however, stiff with respect to constant mean curvature: $\delta K = 0$ for every static source. A static galaxy therefore keeps $K = 3H_0$, and the floor removes MOND from SPARC ($\Delta\chi^2 = +6548/+8030$).

A bound-region switch (L342 6/6, BS1 4/5). One scalar of the foliation, $x = 9R^{(3)}/(4K^2)$, is 0 on FRW, $\frac{3}{2}\Omega_m\delta$ in the linear web and $4\pi G\rho_{\text{dyn}}/H^2$ in a static system. Turning MOND on only for $x > x_c$ returns $\sigma_8 = 0.810$ for $x_c = 5$. Galaxies are untouched (SPARC minimum $x = 596$; Solar neighbourhood 1.2×10^6), and the switch is healthy (induced K^2 coefficient $7.6 \times 10^{-14} \ll c_2$). Scored on the Brouwer et al. KiDS-1000 lensing rotation curves with a retained phantom profile, $x_c \approx 4\text{--}7$ was preferred ($\Delta\chi^2 = -15$ to -19). Whether one threshold serves all four stellar-mass bins is marginal. Free per-bin thresholds $[4, 6, 2, 5]$ improve χ^2 by 8.0/7.5 for 3 parameters ($p = 0.046/0.059$), with no trend with mass. The pre-set $p > 0.05$ gate fails narrowly on the canonical footing. That lead is withdrawn below.

What the switch costs.

- (a) *Planck (L350, 5/5).* On FRW, C-H/K is low-energy Hořava gravity with $\lambda - 1 = c_2$, $\eta = \alpha_c$, $\xi = 1$. Its minisuperspace gives $G_{\text{cos}}/G_N = (2 - \alpha_c)/(2 + 3c_2)$, eq. (9) of Frusciante & Benetti [FB21]. Their Planck-era fits cap c_2 at $6.3 \times 10^{-4}\text{--}2.9 \times 10^{-3}$ (95%), 2.5–11.6 times below the tracking floor, so the window above is excluded. At the caps the phantom still tracks galaxy bodies, but deep-MOND outskirts are amplified by up to 40% and fast ones are left behind; KM2's effect then reaches KiDS precision. A repair is available: the leaf-average-subtracted term $-c_2(K - \langle K \rangle_{\Sigma})^2$ leaves Friedmann's equation GR's and acts on every $k \neq 0$ mode exactly as before. It is a global term on C-H's closed leaves, offered as a construction.
- (b) *GW170817 (L351, 5/5).* Because x contains $R^{(3)}$, the switch rescales only the tensor gradient term: $c_T^2 = 1 + \frac{9}{4}(2\alpha^2 q/K^2)f'(x)$. Across one switching shell the graviton gains exactly $GM_b/(6c^3)$ on light, whatever x_c , f or H : 13.6 h for the Milky Way. GW170817 would have arrived ≥ 18.8 h before the gamma rays, against 1.74 s [LVC17]. The shear-completed variable $\tilde{x} = 9(R^{(3)} + \sigma_{ij}\sigma^{ij})/(4K^2)$ takes the same values on FRW, in the linear web and in static systems, and keeps $c_T = 1$ exactly. C-H/K without the switch already has $c_T = 1$.

- (c) *Gauss (L352, 7/7)*. The phantom is a divergence, so a switched MOND flux makes an isolated galaxy weigh exactly its baryons beyond the switch edge. The cancellation is carried by a negative-mass shell: $-93M_b$ for $M_b = 6 \times 10^{10} M_\odot$ with the edge at 1 Mpc, singular as a static solution. The retained profiles used by L341, L342 and BS1 cannot be produced by the action. Rescored with the realizable profile, $x_c = 5$ is disfavoured ($\Delta\chi^2 = +35/+36$ with a free two-halo term, $+214/+225$ without). KiDS accepts only $x_c \approx 2-3$, and only with bias-like two-halo amplitudes (≤ 2), for which linear growth stays within 2% of Λ CDM.
- (d) *The Lyman- α forest (L346 6/6, L347 4/5)*. In matter power at $k = 1-4 h \text{ Mpc}^{-1}$ and $z = 2-3$, $x_c = 5$ exceeds Λ CDM by 6–36%, failing the pre-declared $[0.8, 1.2]$ band; $x_c = 7$ fails at $z = 2$ only. On the observable, the 1D flux power at fixed mean transmission, the worst deviation is 10.8% at $x_c = 5$ (9.6% canonical) and 7.2% at $x_c = 7$, against a pre-declared 10%. That is borderline, and the margin is below the method’s systematics (FGPA on a $0.2-0.4 h^{-1} \text{ Mpc}$ grid, one realisation). The thresholds KiDS accepts ($x_c \lesssim 3$) have not yet been run on the flux observable. The flux deviation grows as x_c falls (7.2% at 7, 10.8% at 5), so a KiDS–forest pincer is expected but is *not* claimed here. L352’s statement that it is closed rests on matter power.
- (e) *The external field (BS2 11/15, BS3 6/7; the failing checks are the construction’s own gates)*. We use an exact stacked-lens QUMOND flux law with an external field ($M_{\text{dyn}} = M_b N(y, e)$, sphere-averaged, limits verified). Isolated MOND over-predicts KiDS at 1–3 Mpc. The switch fixes this, but a weak external field $e \approx 7 \times 10^{-5}$ does most of the same work ($-13.5/-14.2$), and the data cannot reliably tell the two truncations apart. KiDS bounds a shared Newtonian external field at $e \leq 7.2 \times 10^{-5}/5.2 \times 10^{-5}$. The switch keeps the web Newtonian and Λ CDM-like, so the web pulls on every lens with $\sigma_{g,3D} = 0.0131/0.0108 a_0$ ($\sigma_{v,1D} = 265 \text{ km s}^{-1}$), 194–236 times the stacked bound. C-H/K’s kernel reads free-fall fields: L340 applies it to the Sun in the Galaxy’s field. The result is $\Delta\chi^2 = +569/+582$ on all 60 points and $+40/+45$ inside 0.3 Mpc, where no two-halo term can help. BS2 left two gaps: the field an *isolated* lens feels, and the two-halo term. At the 4 times weaker field Brouwer et al. adopted ($e = 0.003$) the construction passes inside 0.3 Mpc and fails only at 0.3–3 Mpc ($+320/+362$). BS3 (6/7) closes both gaps. In a linear Λ CDM mock (300 Mpc, 256^3 , no neighbour within 3 Mpc), isolated lenses feel rms $|g| = 0.0150 a_0$, the same as random points, because the long modes that dominate the field are shared; only 1% are as quiet as $e = 0.003$. With a two-halo term free per mass bin (bias ≤ 2) given to every model, the construction sits at $\Delta\chi^2 = +404/+415$ on all points and $+47/+52$ inside 0.3 Mpc. Allowing bias up to 20 still leaves $+364/+375$. When the construction is injected as truth, the test passes in 10 of 10 draws. C-H/K plus the switch, with its kernel reading the total Newtonian field of a Λ CDM-like web, is therefore excluded by KiDS-1000 isolated lensing. A kernel sourced by baryons only is disfavoured on all points ($+145/+152$) but survives inside 0.3 Mpc. Pure MOND at its own MOND-level field shows a $+17/+16$ tension at 0.3–3 Mpc only.
- (f) *The dark carrier (L345, 2/2)*. C-H/K’s MOND sector reads the total lapse, so a minimally coupled carrier is boosted exactly like baryons, statically and in motion. For the programme’s Λ -triggered kicked-decay carrier over $f_d(0) = 0.8-0.99$ and $v_k = 1000-3000 \text{ km s}^{-1}$, no cell passes X-COP [Eck19], S_8 and galaxies together.

Net. Without a switch, C-H/K fails linear growth by an order of magnitude. With a switch made consistent with GW170817 ($\tilde{x}$) and Planck (leaf-average term), Gauss’s law leaves at most a KiDS window at $x_c \approx 2-3$, and the construction’s own web field fails KiDS isolated lensing unless the kernel is blind to it. *C-H/K is a local-gravity construction without a working cosmology.*

6 The specification any completion must now meet

- (i) The linear web must stay Newtonian (σ_8), and the $z \approx 2-3$ intergalactic medium nearly so (flux power within $\sim 10\%$).
- (ii) The Mpc-scale lensing flux around isolated galaxies must be supplied. By Gauss’s law that means MOND switched on there, or real mass.
- (iii) Whatever supplies (ii) must not itself be MOND-boosted (L345).
- (iv) The kernel must not read a Λ CDM-level large-scale Newtonian field ($\sim 10^{-2} a_0$; KiDS allows $e \lesssim 7 \times 10^{-5}$), or the web’s field must itself be MOND-level, as in pure MOND.
- (v) Any switch variable must enter the kinetic and gradient terms together ($c_T = 1$).

The doors this leaves are each a new ingredient: a kernel blind to the large-scale field, for example one sourced by the bound region’s own matter (BS3; a baryons-only kernel survives only inside 0.3 Mpc); a carrier coupled directly to U (L345); a time-dependent threshold; and a khronon sector that itself behaves as dust on FRW, the route of Blanchet & Skordis [BS24, BS25]. Running that route through the same two gates is the natural next test; the moving-source half is begun in lane BSK1. One prediction of the local construction survives whatever the

cosmology turns out to be. Stacked weak lensing split by the lens’s velocity through the CMB frame should show the $1/(1 - v_{\parallel}^2/c_s^2)$ amplification, at up to tens of per cent in deep-MOND outskirts if c_2 sits at the Planck caps.

7 What this changes for the core relation

- $a_0 = \kappa c \sqrt{G \rho_{\Lambda}}$ is unaffected as a phenomenological law. Neither test uses or moves κ , which remains fitted (0.465 ± 0.076) and, by the zero-mode theorem, free in local actions [Z26a].
- The one published first-principles route to $a_0 \propto \sqrt{\Lambda}$ reproduces the scaling but not the coefficient (factor 2.8–3.1), the shape, or the matter-tracking. It cannot be cited as a derivation of the relation.
- The relation’s distinctive observational prediction, a_0 flat in redshift (tested by deep-MOND rotators at $z \approx 2.5$ [Z26c]), is independent of both results. If a switch of this kind is used, its lensing edge at $z = 2.5$ lies at 0.3–0.5 Mpc for a Milky-Way mass, far outside the rotation curves that test it.
- The completion question has sharpened. Local gravity with a moving phantom is achievable at the orders computed, and the open problem is cosmology, in the form (i)–(v) above.

8 Corrections carried on the record

Within the week, later lanes corrected earlier ones. The corrections are kept, not overwritten:

- L342’s KiDS preference and L341’s truncation result are withdrawn (L352: the profile they scored is unrealizable). BS2 shows the preference is not specific to the switch in any case.
- L340’s upper c_2 edge (BBN) is superseded and its window excluded by Planck-era fits (L350).
- KM2: the “lag” predicted in L340 is an amplification.
- L330’s frozen-phantom identity is linear around a phantom-free background (§3). In the nonlinear treatment the phantom is conserved and carried by the aether, so its statement that an existing phantom cannot follow a moving source is not established for C-H.
- KM1’s “no velocity-dependent effect” holds for the metric potentials only, not for the MOND sector’s input (KM1 C7, L333).
- L346’s matter-power failure is borderline on the flux observable (L347), and L352’s forest leg rests on matter power (§5d).

9 Reproducibility

All lanes live in the public repository <https://github.com/carlzimmerman/zimmerman-formula>. For this note each was re-run from a clean checkout of commit b48b224c5 (BS3: 910469341), and each reproduced its committed check summary. The KiDS-dependent lanes (L341, L342, L352, BS1–BS3) read the public Brouwer et al. (2021) lensing-profile release, which the repository does not redistribute; it must be placed in `real_research/data/lensing_rar/brouwer2021_rar/`. Each lane has a mutation control, run with `MUTATE=1`, which writes separate outputs and must fail a named load-bearing check.

lane	script (under <code>real_research/</code>)	checks	mutation control (must fail)
L313	<code>cq_gravity_2026/L313_cq_rectification_lensing.py</code>	20/20	drop $\Gamma\Gamma$ terms: Isaacson and Schwarzschild controls fail
L314	<code>cq_gravity_2026/L314_cq_homogeneous_noise_pincer.py</code>	4/4	off-shell background: S1 fails
L330	<code>g03_audit_2026/L330_moving_source_momentum_gate.py</code>	5/5	momentum channel added: frozen identity fails
L333	<code>g03_audit_2026/L333_c2_channel_carries_the_phantom.py</code>	9/9	phantom given the baryons’ momentum: the $1/c_2$ term vanishes
L340	<code>g03_audit_2026/L340_filtered_khronon_completion.py</code>	11/11	ν_{RAR} kernel: A1, H2 fail
KM1	<code>khronon_momentum_2026/KM1_khronon_carries_phantom.py</code>	7/7	$\varepsilon = 0$: C3, C4 fail
KM2	<code>khronon_momentum_2026/KM2_phantom_lag_law.py</code>	4/4	$c_2 = 0$: tracking fails
KM3	<code>khronon_momentum_2026/KM3_chk_one_pn.py</code>	5/5	$\gamma = 2$ exterior gives $\beta = 3/2$
L341	<code>g03_audit_2026/L341_chk_fr_w_gate.py</code>	7/7	MOND off: F2 fails

lane	script (under <code>real_research/</code>)	checks	mutation control (must fail)
L342	<code>g03_audit_2026/L342_bound_region_switch.py</code>	6/6	no switch: B2 fails
L345	<code>dark_sector_2026/L345_chk_universal_carrier_pincer.py</code>	2/2	additive coupling: window reappears
L346	<code>g03_audit_2026/L346_switch_forest_gate.py</code>	6/6	switch never on: gate flips
L347	<code>g03_audit_2026/L347_switch_forest_flux_power.py</code>	4/5 ^a	switch never on: F2, F3 flip
L350	<code>g03_audit_2026/L350_chk_cosmological_G_gate.py</code>	5/5	WMAP7-era bound: G2 fails (window survives)
L351	<code>g03_audit_2026/L351_switch_gw170817_gate.py</code>	5/5	no σ^2 completion: W4 fails
L352	<code>g03_audit_2026/L352_switch_gauss_compensation.py</code>	7/7	retained profile: Z3 flips
BS1	<code>switch_audit_2026/BS1_per_bin_threshold.py</code>	4/5 ^b	drifting synthetic data: S3 fails
BS2	<code>switch_audit_2026/BS2_efe_vs_switch.py</code>	11/15 ^c	EFE-only synthetic data: E3a fails
BS3	<code>switch_audit_2026/BS3_isolated_field_two_halo.py</code>	6/7 ^d	inject the no-EFE model: G3 fails
BSK1	<code>bs_khronon_2026/BSK1_moving_source_gate.py</code>	8/8 ^e	aether pinned: M3, M4 fail

^a The failing check is informational (resolution monotonicity). ^b The pre-set single-threshold gate fails narrowly on the canonical footing. ^c The failures are the construction's own gates (E3b, E6, E9, E13), recorded as results. ^d G1, the construction's own gate, fails: this is the exclusion. ^e New with this note; cited here only for its M2.

The Lean 4 certificates are in `fable_independent_2026/lean_2026/`: `L330_frozen_density`, `L340_chk_certificates`, `L341_frw_certificates`, `L342_switch_certificates`, `I24_c2_channel_phantom`, `I25_chk_universal_coupling`, `I26_switch_forest_threshold`, `KM_khronon_momentum`, and `L350_L352_chk_gate_certificates`. They contain 62 theorems, recompiled for this note with `lake env lean <file>` (Mathlib, exit 0, zero sorry, axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$). They certify algebraic identities: the frozen-density identity, the tracking and lag laws, the Hořava G ratio, the shell advance, the Gauss cancellation, and $x \geq \frac{3}{2}\Omega_m\delta$. They do not certify the physical premises or the numerics.

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